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1 Relative Strength Index

Relative strength is the ratio of average upward price movement to average downward price movement over a selected period:

$$RS = \frac{\text{Average gain}}{\text{Average loss}}.$$

Suppose the relative strength is

$$RS = 2.0.$$

This means that over the measured period, upward price movement has been twice as large as downward price movement.

- $RS > 1$: gains have been larger than losses.
- $RS = 1$: gains and losses have been equal.
- $RS < 1$: losses have been larger than gains.
- $RS \rightarrow \infty$: there have been gains but essentially no losses.
- $RS = 0$: there have been losses but no gains.

The Relative Strength Index, or RSI, converts relative strength into a bounded value between 0 and 100:

$$RSI = 100 - \frac{100}{1 + RS} = 100 \left(\frac{RS}{1 + RS} \right).$$

As $RS \rightarrow \infty$, $RS/(1 + RS) \rightarrow 1$, so $RSI \rightarrow 100$. When $RS = 1$, $RSI = 50$. When $RS \rightarrow 0$, $RSI \rightarrow 0$.

Relative Strength	RSI
0	0
0.25	20
0.5	33.33
1	50
2	66.67
3	75
4	80
Very large	Approaches 100

RSI is commonly interpreted as follows:

- **Above 70:** frequently described as overbought.
- **Below 30:** frequently described as oversold.
- **Above 50:** upward momentum is dominant.
- **Below 50:** downward momentum is dominant.

To calculate the RSI for a time t , we need

$$RSI_t = \frac{\text{AvgGain}_t}{\text{AvgLoss}_t}.$$

For each candlestick c_j , compute the close-to-close change.

$$\Delta_{c_j} = \text{Close}_{c_j} - \text{Close}_{c_{j-1}}.$$

The change is called a gain if it is greater than zero, it is called a loss if it is less than zero.

$$\text{Gain}_{c_j} = \max(\Delta_{c_j}, 0), \quad \text{Loss}_{c_j} = \max(-\Delta_{c_j}, 0).$$

Then, compute (for a 14-period RSI)

$$\text{AvgGain}_{14} = \frac{\sum_{j=1}^{14} \text{Gain}_{c_j}}{14}, \quad \text{AvgLoss}_{14} = \frac{\sum_{j=1}^{14} \text{Loss}_{c_j}}{14}.$$

1.1 Wilder's smoothing

Wilder's smoothing is a recursive moving-average method used in RSI, ATR, and several other indicators. It gives more weight to recent observations while retaining information from all earlier observations.

For a period n , the update formula is

$$S_t = \frac{(n-1)S_{t-1} + X_t}{n}.$$

- X_t is the newest observation.
- S_{t-1} is the previous smoothed value.
- S_t is the new smoothed value.
- n is the smoothing period.

For the relative strength, instead of using

$$\text{AvgGain}_{14} = \frac{\sum_{j=1}^{14} \text{Gain}_{c_j}}{14}, \quad \text{AvgLoss}_{14} = \frac{\sum_{j=1}^{14} \text{Loss}_{c_j}}{14},$$

we use Wilder's smoothing:

$$\begin{aligned} \text{AvgGain}_{c_j} &= \frac{13\text{AvgGain}_{c_{j-1}} + \text{Gain}_{c_j}}{14} \\ \text{AvgLoss}_{c_j} &= \frac{13\text{AvgLoss}_{c_{j-1}} + \text{Loss}_{c_j}}{14}. \end{aligned}$$

Then,

$$RS_{c_j} = \frac{\text{AvgGain}_{c_j}}{\text{AvgLoss}_{c_j}}.$$

Consequently,

$$RSI_{c_j} = 100 \left(\frac{RS_{c_j}}{1 + RS_{c_j}} \right).$$

The recursive formula requires a previous smoothed value, so the first value is normally initialized with a simple average of the first n observations:

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

For the RSI, the initial average gain and initial average loss are simply the averages

$$\text{AvgGain}_{14} = \frac{\sum_{j=1}^{14} \text{Gain}_{c_j}}{14}, \quad \text{AvgLoss}_{14} = \frac{\sum_{j=1}^{14} \text{Loss}_{c_j}}{14}.$$

Every later value uses Wilder's recursive update.

2 Moving averages

A moving average is commonly used with time series data to smooth out short-term fluctuations and highlight longer-term trends or cycles

Given a series of numbers and a fixed subset size, the first element of the moving average is obtained by taking the average of the initial fixed subset of the number series. Then the subset is modified by "shifting forward"; that is, excluding the first number of the series and including the next value in the series

A moving average is commonly used with time series data to smooth out short-term fluctuations and highlight longer-term trends or cycles - in this case the calculation is sometimes called a time average.

2.1 Simple Moving Average

In financial applications a simple moving average (SMA) is the unweighted mean of the previous k

$$\text{SMA}_k = \frac{1}{k} \sum_{i=n-k+1}^n p_i.$$

When calculating the next mean $\text{SMA}_{k,\text{next}}$ with the same sampling width k the range from $n - k + 2$ to $n + 1$ is considered. A new value p_{n+1} comes into the sum and the older value p_{n-k+1} drops out. This simplifies the calculations by reusing the previous mean $\text{SMA}_{k,\text{prev}}$.

$$\begin{aligned} \text{SMA}_{k,\text{next}} &= \frac{1}{k} \sum_{i=n-k+2}^{n+1} p_i \\ &= \frac{1}{k} \left(p_{n-k+2} + p_{n-k+3} + \cdots + p_n + p_{n+1} + \underbrace{p_{n-k+1} - p_{n-k+1}}_0 \right) \\ &= \frac{1}{k} (p_{n-k+1} + p_{n-k+2} + \cdots + p_n) - \frac{p_{n-k+1}}{k} + \frac{p_{n+1}}{k} \\ &= \text{SMA}_{k,\text{prev}} + \frac{1}{k} (p_{n+1} - p_{n-k+1}). \end{aligned}$$

2.2 Weighted Moving Average

A weighted average is an average that has multiplying factors to give different weights to data at different positions in the sample window.

In the financial field, and more specifically in the analyses of financial data, a weighted moving average (WMA) has the specific meaning of weights that decrease in arithmetical progression.[5] In an n -day WMA the latest day has weight n , the second latest $n - 1$, etc., down to one.

$$\text{WMA}_M = \frac{np_M + (n-1)p_{M-1} + \cdots + 2p_{(M-n)+2} + p_{(M-n)+1}}{n + (n-1) + \cdots + 2 + 1}.$$

2.3 Exponential Moving Average

3 VWAP (Volume Weighted Average Price)

The average price a stock has traded at throughout the day weighted by volume. Instead of treating each price the same, VWAP assigns more weight to prices where trading activity was heavier.